

Homework 1

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Due date: October 1, 2026

1. In this problem, you are asked to implement a few linear matrix inequalities (LMIs).

(a) (10 points) Consider the dynamical system $x_{k+1} = Ax_k$, where

$$A = \begin{bmatrix} -0.5 & -0.82 & 0.31 \\ -0.23 & 0.61 & 0.57 \\ 0.31 & -0.11 & -0.36 \end{bmatrix}.$$

Test the stability of this system by implementing the following semidefinite program (SDP):

$$\begin{aligned} P &\succ 0, \\ A^\top P A - P &\prec 0. \end{aligned}$$

Find P to satisfy the above linear matrix inequalities. You are allowed to use AI to help you implement an SDP for this problem. Report the final P you find.

- (b) (15 points) Consider the gradient method $x_{k+1} = x_k - \alpha \nabla f(x_k)$, where f is L -smooth and m -strongly convex. In class, we showed that the gradient method converges at a rate ρ if there exists $\lambda \geq 0$ such that

$$\begin{bmatrix} 1 - \rho^2 & -\alpha \\ -\alpha & \alpha^2 \end{bmatrix} + \lambda \begin{bmatrix} -2mL & m + L \\ m + L & -2 \end{bmatrix} \preceq 0.$$

Test the above matrix inequality numerically. For each pair

$$(m, L) \in \{(1, 10), (1, 100), (0.1, 20), (1, 1000), (1, 5000), (1, 50000)\},$$

try a few values of α satisfying $0 < \alpha < 2/L$, and find the smallest ρ^2 for which the LMI is feasible. Report $1 - \rho$ and λ in scientific notation, together with the solver name and status. Is ρ equal to $\max\{|1 - m\alpha|, |1 - L\alpha|\}$? What happens to the numerical computation when L/m is very large?

2. (25 points) Consider the gradient method $x_{k+1} = x_k - \alpha \nabla f(x_k)$, where f is L -smooth and m -strongly convex, with $0 < m < L$. In class, we showed that the gradient method converges at a rate ρ if there exists $\lambda \geq 0$ such that

$$\begin{bmatrix} 1 - \rho^2 & -\alpha \\ -\alpha & \alpha^2 \end{bmatrix} + \lambda \begin{bmatrix} -2mL & m + L \\ m + L & -2 \end{bmatrix} \preceq 0.$$

Use this condition to show that gradient descent with any constant step size $0 < \alpha < 2/L$ converges at the rate

$$\rho = \max\{|1 - m\alpha|, |1 - L\alpha|\}.$$

Write out the proof. What is the choice of λ ?

3. Consider the three-layer feedforward neural network

$$f(x) = W_3\sigma(W_2\sigma(W_1x + b_1) + b_2) + b_3,$$

where σ is the ReLU activation function applied entrywise, $W_i \in \mathbb{R}^{n_i \times n_{i-1}}$, and $b_i \in \mathbb{R}^{n_i}$ for $i = 1, 2, 3$. All vector norms are Euclidean.

- (a) (25 points) Prove the following LipSDP condition from Lecture 9: if there exist $L \geq 0$ and positive definite diagonal matrices $T_1 \in \mathbb{R}^{n_1 \times n_1}$ and $T_2 \in \mathbb{R}^{n_2 \times n_2}$ such that

$$\begin{bmatrix} -L^2I & W_1^\top T_1 & 0 \\ T_1 W_1 & -2T_1 & W_2^\top T_2 \\ 0 & T_2 W_2 & W_3^\top W_3 - 2T_2 \end{bmatrix} \preceq 0, \quad (1)$$

then f is L -Lipschitz, i.e.,

$$\|f(x) - f(y)\| \leq L\|x - y\| \quad \text{for all } x, y \in \mathbb{R}^{n_0}.$$

Write out a detailed proof using the incremental quadratic constraints for the two ReLU layers. You may use the fact that for any positive definite diagonal matrix T of the appropriate size,

$$\begin{bmatrix} u - v \\ \sigma(u) - \sigma(v) \end{bmatrix}^\top \begin{bmatrix} 0 & T \\ T & -2T \end{bmatrix} \begin{bmatrix} u - v \\ \sigma(u) - \sigma(v) \end{bmatrix} \geq 0 \quad \text{for all } u, v.$$

(b) (25 points) **Numerical experiment.** Consider the three-layer ReLU network

$$f_a(x) = W_3 \sigma(W_2 \sigma(W_1 x)),$$

with zero biases and

$$W_1 = \begin{bmatrix} 1 & 1 \\ a & -a \end{bmatrix}, \quad W_2 = \begin{bmatrix} a & 1 \\ -a & 1 \end{bmatrix}, \quad W_3 = \begin{bmatrix} 1 & -1 \end{bmatrix}.$$

For each $a \in \{1, 2, 5, 10\}$, carry out the following steps.

(i) Compute the spectral norm product bound

$$L_{\text{prod}} = \|W_3\|_2 \|W_2\|_2 \|W_1\|_2,$$

where $\|W\|_2$ is the largest singular value of W .

- (ii) Implement the LipSDP condition (1). Replace L^2 by a scalar decision variable $\zeta \geq 0$ and minimize ζ over diagonal matrices $T_1, T_2 \succeq 10^{-6}I$ subject to the LMI. Submit your code and report $L_{\text{SDP}} = \sqrt{\zeta^*}$, the solver name, and its status. Let M^* denote the matrix on the left-hand side of (1), evaluated at your unrounded solver output with $L^2 = \zeta^*$. Check numerical LMI feasibility by verifying $\lambda_{\max}(M^*) \leq 10^{-5}$. You may use AI to help implement the SDP.
- (iii) Tabulate the two bounds and their ratio $L_{\text{prod}}/L_{\text{SDP}}$, with all three values rounded to four decimal places. Add a column for $\lambda_{\max}(M^*)$ in scientific notation. How does the improvement change as a increases?

Grading note: The optimal multipliers T_1, T_2 need not be unique, and their individual entries are not required. Any pair satisfying the constraints to numerical tolerance and attaining the optimal bound is acceptable.

Optional: Evaluate

$$\frac{\|f_a(x) - f_a(y)\|}{\|x - y\|} \quad \text{at } x = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad y = \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

Compare this lower bound on the true Lipschitz constant with the two upper bounds. What does this suggest about the tightness of LipSDP for this example?